Rate-controllable PPG synthesis for a low-cost simulator

Command

rate 40-180 bpm
irregularity
recording context

Rhythm scaffold

beat times fixed
from the command

Learned waveform

PhysioMix: GAN, 0.16 M
PhysioFlow: rectified flow, 6.9 M

Raspberry Pi 4

2 x 12-bit DAC at 500 Hz
red and infrared LED driver
optical path not yet validated

Rate error below 1 bpm from 40 to 180 bpm

Pre-registered selection rules

Unseen subjects rated indistinguishable: PhysioFlow 50-91 %, PhysioMix 7-45 %, parametric 0 %

Everyday wrist PPG from an unseen device: below 1 % for both models